Trigonometry

Section 1: Unit Circle & Special Angles

1 What is the exact value of $\sin(150^\circ)$?

a. $\frac{1}{2}$

b. $-\frac{1}{2}$

c. $\frac{\sqrt{3}}{2}$

d. $-\frac{\sqrt{3}}{2}$

2 What is the exact value of $\cos(240^\circ)$?

a. $-\frac{1}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $\frac{\sqrt{3}}{2}$

3 Convert $\frac{5\pi}{6}$ radians to degrees.

a. 150°

b. 120°

c. 135°

d. 210°

Section 2: Pythagorean Identities & Simplifications

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4 Simplify the trigonometric expression: $\frac{\sin^2\theta}{1-\cos^2\theta}$.

a. 1

b. $\tan^2\theta$

c. $\cot^2\theta$

d. $\sin\theta$

5 If $\cos\theta = \frac{3}{5}$ and θ is an acute angle in Quadrant I, what is $\tan\theta$?

a. $\frac{4}{3}$

b. $\frac{3}{4}$

c. $\frac{4}{5}$

d. $\frac{5}{3}$

6 What is the simplified form of $\sec\theta\cot\theta$?

a. $\csc\theta$

b. $\sin\theta$

c. $\cos\theta$

d. $\tan\theta$

Section 3: Double Angle Formulas & Equations

7 If $\sin\theta = \frac{3}{5}$ and $\cos\theta = \frac{4}{5}$, what is the value of $\sin(2\theta)$?

a. $\frac{24}{25}$

b. $\frac{12}{25}$

c. $\frac{7}{25}$

d. $\frac{18}{25}$

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8 What are the solutions to $2\cos\theta - 1 = 0$ on the interval $[0, 2\pi)$?

a. $\frac{\pi}{3}, \frac{5\pi}{3}$

b. $\frac{\pi}{6}, \frac{11\pi}{6}$

c. $\frac{2\pi}{3}, \frac{4\pi}{3}$

d. $\frac{\pi}{4}, \frac{7\pi}{4}$

9 Which expression is equivalent to $\cos(2\theta)$?

a. $\cos^2\theta - \sin^2\theta$

b. $2\cos^2\theta + 1$

c. $1 - 2\cos^2\theta$

d. $\cos^2\theta + \sin^2\theta$

Section 4: Laws of Sines & Cosines

10 In $\triangle ABC$, $a = 8$, $\sin A = 0.4$, and $\sin B = 0.6$. What is the length of side b?

a. 12

b. 10

c. 14

d. 16

11 In $\triangle ABC$, $b = 5$, $c = 6$, and $\angle A = 60^\circ$. What is the value of a^2?

a. 31

b. 61

c. 41

d. 25

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12 What is the area of a triangle with side lengths 6 cm and 8 cm enclosing an angle of 30°?

a. 12 cm^2

b. 24 cm^2

c. $12\sqrt{3}$ cm^2

d. 16 cm^2

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